

Computable Countermarkets and the Limits of Universal Trading

Karl Svozil ¹

¹*Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8-10/136, A-1040 Vienna, Austria**
(Dated: September 19, 2026)

Every universal-trading claim pairs a trader with a market—a path, generator, or law. For any total deterministic computable trader, its code yields a fixed positive rational countermarket for one discounted asset: every proportional return is either $+\epsilon$ or $-\epsilon$, every active position loses, and inactivity earns zero. Thus the obstruction persists when flat paths are excluded. The model is frictionless and self-financing, with rational orders and no computation deadline. Gold-style learning cannot identify every computable binary market rule from history. Separately, Turing-universal generators make certification of unbounded-future events undecidable; Busy-Beaver growth defeats every computable description-size waiting schedule. A computable process law produces Martin-Löf-random records almost surely, while finite observations cannot certify randomness; no-arbitrage supplies a distinct financial boundary. Together these results form an expository taxonomy: repeatable success requires a market restriction, benchmark, risk premium, or informational advantage, while time reversal and within-class permutations supply simple stress tests.

I. INTRODUCTION

Throughout this paper the elementary configuration has two sides: a *trader* and a *market*. At each date the trader converts observed history into a position; the market supplies the next price. Depending on the question, the market side is a realized path, a path-generating program, a probability law, or a price process constrained by no-arbitrage. Holding the trader fixed while changing the admitted market side reveals which universal claim is at issue.

A strategy is *universal* here only if, after discounting by a specified numeraire, it produces strictly positive terminal gain from zero initial capital on every admitted trajectory at a specified finite horizon. This differs from non-negative gain everywhere with positive gain somewhere, eventual profit at a path-dependent date, expected profit under a law, and small regret to a benchmark. Zero cost, self-financing, admissibility, and discounted gain distinguish this claim from positive wealth obtained by investing positive capital. Its quantifier order is

$$\exists \sigma \forall \mathcal{E} : \quad \sigma \text{ wins against } \mathcal{E}, \quad (1)$$

whereas diagonalization establishes

$$\forall \sigma \exists \mathcal{E}_\sigma : \quad \sigma \text{ does not win against } \mathcal{E}_\sigma. \quad (2)$$

For a fixed total deterministic computable trader, its code can be embedded in a total program that recursively chooses the next bounded proportional price move opposite to each nonzero position and a nonzero move during inactivity. The resulting countermarket is one fixed computable path; any finite prefix can be generated offline. A trader that never takes a position earns zero, which already refutes a guarantee of strict profit. This is the trading counterpart of the next-bit construction

$x_0 = 1 - F(\emptyset)$ and $x_t = 1 - F(x_0, \dots, x_{t-1})$ for $t \geq 1$: it defeats the fixed proposed rule, not every rule on one exceptional sequence.

Three complementary limits use different market information. Gold’s learner sees an accumulating history and must eventually stabilize on a correct program; no effective learner identifies every binary total recursive time function, although restricted classes such as the primitive-recursive time functions are identifiable in the limit [1, 2]. A source-code procedure that settled every future event of Turing-universal generators would instead decide the Halting Problem, while Busy-Beaver growth excludes a computable worst-case waiting horizon. Finally, relative to a specified computable measure, a Martin-Löf-random record escapes every effective null test. Such records have measure one, but finite samples cannot certify this infinite-sequence property. No-arbitrage supplies a separate pricing-measure boundary.

Repeatable practical success must therefore identify structure absent from the unrestricted market class: persistence, mean reversion, a risk premium, a benchmark, lawful information, or another distributional asymmetry. Large, crowded, or public strategies may also change the distribution through price impact and adaptation. This falsifiable restriction on the market side is the operational meaning of “insight” used below.

The explicit diagonal countermarket is the constructive core of the paper. Table I organizes the complementary limits into a taxonomy that keeps distinct several statements commonly summarized as “no universal strategy.” The construction and its proofs are expository reformulations rather than a claim of a newly discovered impossibility theorem. The analysis connects this quantifier map to the literature on unpredictable prices [3–5], Gold learning, computability and randomness, no-arbitrage and NFL, time-reversal and permutation stress testing, the Wheel, Cover universality, and conditional FAPP strategies.

* karl.svozil@tuwien.ac.at; <http://tph.tuwien.ac.at/~svozil>

TABLE I. Distinct claims organized by the trader–market configuration. They are complementary, not interchangeable.

Mechanism	Market side	Formal scope	Conclusion
Diagonal countermarket	Fixed total program constructed from the fixed trader’s code	Pathwise worst case: $\forall \sigma \exists P_\sigma$	No total deterministic computable trader wins on every computable price path
Gold identification in the limit	Accumulating computable history, or interaction with a black box	Success is eventual stabilization on a correct program, without a known stopping date	No effective learner identifies every computable binary time function; restricted classes can be identified
Halting reduction and Busy Beaver	Total program or generator, under an explicit promise	Individual encoded computation over an unbounded horizon	No general computable profit or market-event decider, or computable description-size waiting horizon
Algorithmic randomness	Passive binary record under the computable fair-coin reference law	Each Martin-Löf-random path; such paths have measure one	Independent fair trials yield ML-random records almost surely; finite tests cannot certify randomness
No-arbitrage	Stochastic price process admitting a coherent pricing measure	Almost-sure payoff conditions under an ELMM	No admissible zero-cost classical arbitrage
Cover universality	Passive bounded sequence and a stated benchmark class	Pathwise, but benchmark-relative and asymptotic	Vanishing per-period regret, not guaranteed positive profit

II. THE TRADER AND THE MARKET: COMPUTABILITY AND RANDOMNESS

We now formalize the trader–market pair. The trader is a program M_σ mapping each observed finite price history to a position; the market side supplies the price history. Four specifications lead to four different boundaries: a fixed computable countermarket tailored to a trader defeats its claim of pathwise universality; Gold’s history protocol precludes identifying every computable rule in the limit; a computationally universal class of market generators makes uniform certification of unbounded future events undecidable; and a computable probability law links process randomness to individual random records through an almost-sure statement.

A. A Computable Countermarket for Every Trader

Throughout the discrete constructions, $P_t = S_t/B_t$ is the discounted price of one asset, where S_t is its price in a fixed currency and $B_t > 0$ is the value of one unit of the chosen risk-free bank account. For example, if both initially equal 100, and later $S_1 = 121$ and $B_1 = 110$, the discounted price rises from 1 to 1.1. This measures performance relative to the bank account, not inflation. The position w_t is a number of shares chosen before P_{t+1} is observed. Trading is self-financing: changes in holdings are funded through the existing share and bank balances, with no external cash injections. More explicitly, holding w_t shares and the residual $V_t - w_t P_t$ in the discounted bank account gives

$$V_{t+1} = V_t + w_t(P_{t+1} - P_t).$$

We omit transaction costs, borrowing limits, leverage caps, and solvency constraints. The counterexample therefore also applies to any subclass of these strategies satisfying additional restrictions.

Let $M_\sigma: \mathbb{Q}_{\geq 0}^{\leq \omega} \rightarrow \mathbb{Q}$ be a total computable function on finite rational histories, encoded in the usual effective way, with $w_t = M_\sigma(P_0, \dots, P_t)$. Totality requires an action in finite computation time on every input; it supplies no prescribed running-time bound or guarantee of meeting a trading deadline. Rational outputs model finite-precision orders and, crucially, make comparison with zero decidable. Let $\mathcal{P}_{\text{comp}}^+$ denote the positive rational-valued computable price paths.

Theorem 1 (Computable Diagonal Countermarket). *Fix rational $P_0 > 0$ and $0 < \epsilon < 1$. For every total deterministic computable strategy M_σ , there exists a fixed path $P^{M_\sigma} \in \mathcal{P}_{\text{comp}}^+$, starting at P_0 , with $P_{t+1}/P_t \in \{1-\epsilon, 1+\epsilon\}$ at every date, on which its self-financing gain*

$$G_T = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t)$$

is non-positive at every finite horizon T . It is strictly negative after the strategy has taken a nonzero position at least once.

Proof. Choose rational $P_0 > 0$ and rational $0 < \epsilon < 1$. Recursively compute $w_t = M_\sigma(P_0, \dots, P_t)$ and set

$$P_{t+1} = \begin{cases} P_t(1-\epsilon), & w_t > 0, \\ P_t(1+\epsilon), & w_t < 0, \\ P_t(1+\epsilon(-1)^t), & w_t = 0. \end{cases} \quad (3)$$

Because M_σ is total and rational-valued, this recursion is a total program for one fixed infinite price path. Its

prices remain positive and computable, and every return has magnitude ϵ . In each period, $w_t(P_{t+1} - P_t) \leq 0$, with strict inequality when $w_t \neq 0$. Summing proves the claim at every finite horizon. $\square$

In the last case of Eq. (3), the alternating market move is a convenient but arbitrary choice. Since $w_t = 0$, any market move yields zero gain in that period. Within the theorem’s fixed-magnitude class, any computable choice between returns $+\epsilon$ and $-\epsilon$ is admissible, allowing a wide variety of market-move sequences in place of the alternating convention.

For a concrete example, fix the rule that holds one share for the first period, shorts one share for the second, and then holds no shares. With $P_0 = 100$ and $\epsilon = 1/10$, Eq. (3) produces

$$(w_0, w_1, w_2) = (1, -1, 0), \\ (P_0, P_1, P_2, P_3) = (100, 90, 99, 108.9).$$

The successive discounted gains are $-10, -9, 0$, so $G_3 = -19$, all in bank-account units. The prices are constructed after fixing the rule; they are illustrative, not market data.

Remark 1 (Logical and Economic Scope of the Countermarket). Theorem 1 uses the trader’s *code*, not a physical market’s observation of a deployed order. Once M_σ is fixed, the countermarket program is fixed and can generate any desired finite prefix offline. It is therefore a lawful, deterministic, computable counterexample, although not necessarily an efficient one computationally. If the market side is regarded as an opponent whose objective is to prevent this trader’s gain, the rule is rational relative to that objective. It is not thereby shown to be an economic equilibrium or an arbitrage-free price process, and another trader may profit on it. Such a trader has, in turn, its own computable countermarket.

This is a direct strategy-relative diagonal construction. The proposed universal trader is used to compute an object outside its success set: long is paired with down, short with up, and neutrality with an alternating nonzero return. Just as the complement sequence defeats a purported predictor of every computable sequence, P^{M_σ} defeats the purported trader on every horizon. Enumeration of all programs is unnecessary because the universal claim itself supplies the program to be diagonalized against. The quantifiers are $\forall M_\sigma \exists P^{M_\sigma}$. A constant discounted price path already gives every self-financing rule zero trading gain and refutes strict profit on a class containing that path. The stronger content of Theorem 1 is that every nonzero position of the selected trader incurs a strict loss even on the class with fixed return magnitude ϵ , which contains no flat path. It does not supply a path on which every strategy strictly loses: permanently long and permanently short positions cannot both lose on the same one-step price move.

B. Learning a Market Rule from History: Gold’s Limit

A trader normally receives neither the market generator’s source code nor a proof of its law. The trader sees an expanding record $x_0, x_1, \dots$ and must infer from each finite prefix how it will continue. Gold’s function-identification model formalizes exactly this information flow. At time n , an effective learner L maps $(x_0, \dots, x_n)$ to the index of a Turing machine. It *identifies x in the limit* if, from some finite date onward, every guess is the same index and that machine computes the whole sequence. The learner need not know when its last change of mind has occurred [1, 2].

Let REC_2 denote the total computable binary sequences. Gold’s negative theorem for binary-valued time functions has the trader–market quantifier order, with L ranging over effective learners,

$$\forall L \exists x \in \text{REC}_2 : \\ L(x_0, \dots, x_n) \text{ does not stabilize} \quad (4) \\ \text{on a program for } x.$$

The adverse history depends on the learner. This is closely analogous to $\forall M_\sigma \exists P^{M_\sigma}$, although Gold’s failure criterion is failure to converge to a fixed correct program name, not financial loss.

The corresponding claim about literal next-move forecasting is sharper and has the short direct diagonal proof already used above. If a total computable predictor F outputs one bit from every finite binary history, the recursive definition

$$x_0 = 1 - F(\emptyset), \quad x_t = 1 - F(x_0, \dots, x_{t-1}) \quad (t \geq 1)$$

produces a computable market record on which F is wrong at every date. Thus the proposed statement—every total next-move predictor has a computable countermarket—is correct, but it is this direct prediction diagonalization, not the literal statement of Gold’s identification theorem. Nonconvergence of program names alone does not count the learner’s next-step errors.

The opposite-bit construction also appears explicitly in Legg’s analysis of universal prediction [6]. The qualifier *computable* is essential. Solomonoff induction defines an uncomputable universal predictive ideal; for computable probabilistic environments its conditional predictions converge, with quantitative error bounds developed by Hutter [7–10]. It therefore evades, rather than contradicts, the diagonal result: it is not a total executable trading rule, and predictive convergence is not Gold’s requirement of stabilizing on one exact generating program. For probability-valued forecasts, V’yugin gives a further, distinct impossibility: a randomized construction can make any partial computable forecaster fail calibration with probability arbitrarily close to one [11]. Calibration, next-bit error, and identification in the limit are three different criteria.

Gold’s results also show why the restriction on the market side matters. Every uniformly effective indexed family of total computable functions is identifiable in the limit: a learner selects the least index consistent with the observed prefix, and eventually eliminates every smaller incorrect index. In particular, primitive-recursive time functions form such a family. In Gold’s interactive black-box model, where the eventual hypothesis must reproduce future input–output behavior, finite-state boxes are weakly identifiable in the limit, whereas the class of primitive-recursive black boxes is not [1]. These are different object classes, not contradictory claims. Gold’s 1967 language model further makes explicit that learnability depends on the admitted class, the presentation of the evidence, and the allowed names [2]. For trading, a forecast can therefore be justified only relative to a specified restriction on possible markets.

C. Universal Computation and the Halting Barrier

Gold’s learner works from observations and may revise its hypothesis indefinitely. The next results ask a separate, stronger operational question: can one terminating program inspect arbitrary encoded traders or market generators and correctly decide their unbounded future behavior? The preceding diagonal construction already disproves a trader that wins on every computable path and does not invoke the Halting Problem. The source-code event-certification question does.

Theorem 2 (Undecidability of Eventual Profit). *Fix the positive computable price path $P_t = t + 1$, $t \in \mathbb{N}$. There is no algorithm which, under the promise that its input code computes a total trading program M_σ , correctly decides for every such code whether*

$$\exists T \geq 1 : \quad G_T(M_\sigma; P) = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t) > 0.$$

Proof. Assume such a profit certifier exists. Given an arbitrary Turing machine A and input x , construct a trading program $M_{A,x}$ as follows. At date t , simulate $A(x)$ for $t + 1$ steps. If it has halted within those steps, set $w_t = 1$; otherwise set $w_t = 0$. Every date requires only a finite simulation, so $M_{A,x}$ is total. Along $P_t = t + 1$, every long position earns one unit in the next period. Hence $G_T > 0$ for some T if and only if $A(x)$ eventually halts. A certifier for eventual trading profit would therefore decide the Halting Problem, which is impossible [12]. $\square$

The reduction can also be placed directly on the market side, which is the form most closely connected to source-code event certification.

Theorem 3 (Undecidability of a Market Event). *There is no algorithm which, under the promise that its input code computes a total positive-price generator G , correctly decides for every such code whether its generated path P^G ever satisfies $P_t^G > P_0^G$.*

Proof. Given a Turing machine A and input x , define the total generator by $P_0^{A,x} = 1$ and, for $t \geq 1$,

$$P_t^{A,x} = 1 + \mathbf{1}\{A(x) \text{ halts within } t \text{ steps}\}, \quad t \geq 1. \quad (5)$$

Each quoted price requires only a finite t -step simulation, so the generator is total and computable. Its initial price is one, and it ever quotes a higher price if and only if $A(x)$ halts. A universal decider for the stated market event would therefore decide the Halting Problem. $\square$

Theorem 3 makes “universal computation” precise: the admissible generators can emulate an arbitrary machine and expose its outcome through a price event. No algorithm can then certify every future event of every such market. Both reductions concern promise domains: the putative decider must halt and be correct on every promised total input. Positive instances are semidecidable by simulation until the event occurs; there is no terminating general procedure for certifying its perpetual absence.

This is not Gold’s theorem. The reduction receives a source code and must halt with a yes-or-no answer; Gold’s learner receives accumulating data and need only stabilize eventually. Indeed, the paths in Eq. (5) are identifiable in the limit: guess the flat path, then switch permanently if the price becomes two. No finite date, however, certifies that the initial guess is final. Thus halting excludes uniform certification, whereas Gold excludes identification in the limit for the class of all binary total recursive time functions. Restricted model classes may still be learnable [13–17].

a. Busy-Beaver horizons: finite glimpses and non-computable waiting Fix a step-counted universal machine V with an effective compiler that adds only fixed overhead for a given program schema, and define the run-time Busy Beaver function by

$$\text{BB}(n) = \max\{\text{time}_V(p) : |p| \leq n \text{ and } V(p) \text{ halts}\}, \quad (6)$$

with value zero if the set is empty. The input n is a program-description budget, not a date; $\text{BB}(n)$ is a run-time, which becomes a possible trading date when the computation is embedded in a market generator. A small increase in rule length may therefore require waiting beyond every computably prescribed scale.

Each $\text{BB}(n)$ is finite, and selected low values can sometimes be proved by exhaustive simulation plus proofs that the remaining programs do not halt. This is not uniform “computability for small arguments”: any finite table can be hard-coded, and its apparent pattern licenses no extrapolation. If a total computable h upper-bounded BB , simulating every program of length at most n for $h(n)$ steps would decide halting. Eventual domination uses a further program-schema argument. For each n , a program of $O(\log n)$ bits can encode n , compute $h(n)$, execute more than $h(n)$ dummy steps, and halt. For all sufficiently large n , its compiled V -code has length at most

n , so $\text{BB}(n) > h(n)$. Thus, up to the fixed machine convention, BB eventually exceeds every total computable function [18, 19].

Calude, Dinneen, and Shu illustrate the finite/infinite divide. For one explicit universal self-delimiting register machine, computation and mathematical proof settled halting through 84-bit programs and certified 64 initial bits of its halting probability Ω_U . They did not compute Busy Beaver values, and the work needed to recover the relevant halting programs from longer supplied prefixes of Ω_U outruns every computable bound. Their result is a formidable finite glimpse, not a scalable computation of an uncomputable object [20].

Applied to Eq. (5), a short generator can quote $P_t = 1$ for a Busy-Beaver-scale interval and then quote $P_t = 2$. Across encoded programs of length at most n , the last halt can occur at $\text{BB}(n)$, up to fixed wrapper overhead. Knowing that horizon would make continued silence a certificate of no later move. Equivalently, with TotComp denoting total computable size-dependent schedules,

$$\forall h \in \text{TotComp} \exists p : \begin{aligned} &V(p) \text{ halts,} \\ &\text{time}_V(p) > h(|p|). \end{aligned} \quad (7)$$

If c is the wrapper's fixed description-length overhead, it is absorbed by the computable envelope $H(n) = \max_{k \leq n+c} h(k)$. Every quote at a specified finite date remains computable by bounded simulation; the generator neither computes Busy Beaver nor selects its maximizer uniformly. Noncomputability lies in certification across the family. Each realized path is deterministic and computable, hence not Martin-Löf random under fair-coin coding.

This barrier is stronger than sensitive dependence or visual complexity alone in a precise decision-theoretic sense. A finite contractive iterated-function system or chaotic map with computable data can generate an intricate phenotype from a short rule, yet Turing universality additionally creates questions about the unbounded future for which no terminating general algorithm exists. The categories can overlap when a dynamical system performs universal computation. This is a worst-case claim across generator descriptions, not evidence that typical financial markets exhibit Busy-Beaver delays [17, 21, 22].

These obstructions concern unbounded horizons. At any specified finite horizon, total programs can still be run and their realized prices and gains computed.

D. Worst-Case Value and Private Randomization

Returning to the direct countermarket, its finite-horizon value can be stated explicitly, and the role of a trader's hidden randomization can be separated from deterministic pathwise opposition.

Corollary 4 (Maximin Value). *At every finite horizon T , if the neutral strategy is allowed, then*

$$\sup_{\text{total computable } M_\sigma} \inf_{P \in \mathcal{P}_{\text{comp}}^+} G_T(M_\sigma, P) = 0.$$

Proof. The neutral strategy guarantees zero, while Theorem 1 gives every strategy a computable countermarket with gain at most zero. $\square$

a. Two sources of randomness Randomness can enter on either side: *private trader randomization* randomly selects positions, while *market randomness* generates price movements. They can occur separately or together and need not have the same effect on profits. Private randomization need not destroy an existing edge, and random market movements can have positive expected discounted returns. The result below uses the stronger assumption that the next discounted price change has conditional mean zero, even given the trader's private seed. Under integrability this gives zero expected gain at each fixed horizon, not convergence of realized profits to zero in the long run.

Proposition 5 (Private Randomization Does Not Create a Guarantee). *Fix a finite horizon T , a rational $P_0 > 0$, and a rational $\epsilon \in (0, 1)$. There is a single passive computable market law, independent of the trader, under which every predictable randomized strategy with integrable gains has expected gain zero. Specifically, let*

$$P_{t+1} = P_t(1 + \epsilon \xi_{t+1}), \quad t = 0, \dots, T-1,$$

where $(\xi_t)_{t \geq 1}$ are independent fair signs, also independent of the trader's private random seed. If w_t is measurable with respect to the information available when the position is chosen and $w_t(P_{t+1} - P_t)$ is integrable, then

$$G_n = \sum_{t=0}^{n-1} w_t(P_{t+1} - P_t), \quad n = 0, \dots, T,$$

is a martingale. In particular, $\mathbb{E}[G_T] = 0$.

Proof. Let R denote the trader's private seed and set $\mathcal{G}_t = \sigma(P_0, \dots, P_t, R)$. Allowing $\mathcal{G}_t$ to contain all of R only strengthens the claim. Predictability makes w_t $\mathcal{G}_t$ -measurable, while ξ_{t+1} is independent of $\mathcal{G}_t$ and has mean zero. Hence

$$\mathbb{E}[w_t(P_{t+1} - P_t) \mid \mathcal{G}_t] = \epsilon P_t w_t \mathbb{E}[\xi_{t+1} \mid \mathcal{G}_t] = 0.$$

The integrability assumption therefore makes $(G_n)_{n=0}^T$ a martingale, and $\mathbb{E}[G_T] = \mathbb{E}[G_0] = 0$. $\square$

Private randomization prevents a deterministic countermarket based only on the public history from mechanically opposing the unseen current draw, but it does not yield strictly positive expected gain in every environment. More precisely, under the proposition's integrability assumptions it excludes $\forall P : \mathbb{P}_R(G_T(P, R) > 0) = 1$ at the fixed horizon T . Each of the finitely many fair-sign prefixes has positive probability and a computable infinite continuation; such a guarantee would imply $G_T > 0$ almost surely jointly, hence $\mathbb{E}[G_T] > 0$, a contradiction. It does not exclude a high win probability, profit under another law, or benchmark-relative success.

b. Optional online-game reading The recursion in Theorem 1 also defines a finite perfect-information game: the trader announces w_t , the environment chooses the next return, and Corollary 4 gives the trader-first maximin value zero. This reading is optional because the fixed deterministic trader's countermarket program already generates the same path offline. Borel determinacy concerns infinite games with Borel payoff sets and does not by itself identify a winner here [23]. Hidden private randomization is a different information structure: Proposition 5 proves zero expected gain under one passive computable fair-sign law, not a mixed-game value without specified strategy spaces.

E. A Passive Market: Martin-Löf Randomness

a. When does process randomness yield a random record? Process randomness describes how individual bits are generated and depend on their past. Martin-Löf (ML) randomness is a property of the resulting *infinite sequence* relative to a specified probability law. *Independent fair bits produce a fair-coin ML-random sequence with probability one.* Equivalently, the sufficient process condition is

$$\mathbb{P}(X_{n+1} = 1 \mid X_1, \dots, X_n) = \frac{1}{2} \quad \text{a.s. for every } n \geq 0, \quad (8)$$

including the first bit at $n = 0$.

Independence alone is insufficient for *fair-coin* randomness: independent bits of fixed bias $p \neq 1/2$ almost surely have limiting frequency p . Fair one-bit probabilities alone are also insufficient: repeating one fair bit forever gives fair marginals but a constant, predictable record. Independence is not necessary for ML randomness relative to the process's *own* computable law, which may include bias or dependence.

For FAPP use, the conclusion is therefore conditional on a justified source model, for example independent fair trials. Tests for bias and serial dependence can challenge that model over observed scales; passing them cannot certify independence indefinitely or the ML randomness of an infinite output. An almost-sure conclusion allows exceptional records of probability zero. A fixed deterministic pseudorandom generator with a finite seed cannot produce a fair-coin ML-random infinite sequence, however well its finite outputs test.

Here is the precise classical result. A computable law μ has cylinder probabilities $\mu([s])$ uniformly approximable to any requested accuracy, where $[s]$ comprises sequences beginning with s . An ML test is a uniformly computable enumerable sequence of open sets (U_k) with $\mu(U_k) \leq 2^{-k}$. A record is μ -ML-random if it avoids $\bigcap_k U_k$ for every such test.

Proposition 6 (Classical Measure-One Result, Martin-Löf). *If a binary process has computable joint law μ , then*

$$\mathbb{P}((X_1, X_2, \dots) \in \text{MLR}_\mu) = 1. \quad (9)$$

This includes independent fair trials, computable Bernoulli biases, and computable dependent laws [24, Sec. IV].

Proof. Each test's failure set has measure zero. There are countably many effective tests, so their union still has measure zero. $\square$

The law must be specified: randomness relative to an atomic law need not mean fresh uncertainty at each trial. For example, either constant record is random for the law that repeats one fair bit forever.

b. Borel normality is weaker A sequence is Borel normal in base b when every word of length k has limiting frequency b^{-k} , for every $k \geq 1$. ML randomness for uniform base- b digits implies normality, but the converse fails. Champernowne's decimal 0.123456789101112..., formed by concatenating the positive integers, is normal in base ten yet computable, hence not ML-random for the uniform decimal law [25]. Normality is difficult to establish for a given infinite record: one must control all block lengths in the infinite limit. No finite frequency check proves it, since any finite prefix admits both normal and nonnormal continuations. A proof from a construction, as for Champernowne, can establish normality; extensive sampling alone cannot.

F. Ramsey Regularity and the Finite-Pattern Trap

A large market record can be pattern-rich even when it contains no exploitable law. Ramsey theory makes this finite combinatorial point without assuming a probability model. For fixed positive integers c, m, q , there is a finite number $R_q(m; c)$ such that every c -coloring of the q -element subsets of any set of size at least $R_q(m; c)$ has an m -element subset whose q -subsets all receive the same color. Thus every red-blue coloring of the pairs among six objects contains a monochromatic triangle. Van der Waerden's theorem similarly forces a monochromatic arithmetic progression of any prescribed finite length in every sufficiently long finite coloring [26, 27]. These thresholds may be enormous, but they are finite and computable, unlike a Busy-Beaver horizon.

The guarantee depends on the coding and is weaker than a trading signal. In a discretized return record, equal symbols at equally spaced dates need not be consecutive, form a price trend, have a favorable sign, or persist beyond the sample. Ramsey existence therefore asserts neither statistical dependence nor causation, predictability, or profit. Nor does it conflict with algorithmic randomness. A fair-coin Martin-Löf-random sequence is Borel normal, so every finite binary word, including arbitrarily long runs and finite chart motifs, occurs with its fair-coin limiting frequency. Randomness here means escaping every effective null test, not local patternlessness [28, 29].

Calude and Longo call regularities forced by data volume *Ramsey-type correlations*. Their incompressibility

argument emphasizes that almost all sufficiently long finite strings resist compression by any fixed positive fraction while still containing the finite regularities forced by Ramsey theory [30]. This is an epistemic warning, not a theorem that every sample correlation is false: a Ramsey configuration is not itself a correlation coefficient, and a pattern observed in one finite history is compatible with both random and computable continuations.

Trading research compounds this pattern abundance with selection. If all M null hypotheses are true, their rejection events are independent, and each has rejection probability exactly α , the probability of at least one false rejection is $1 - (1 - \alpha)^M$; if their probabilities are at most α , this is an upper bound. Without independence, the union bound gives $\min\{1, M\alpha\}$ as an upper bound. This probabilistic multiple-testing problem is distinct from Ramsey's deterministic existence theorem, but its practical lesson is aligned: a trend or cross-market relation retained after searching many assets, windows, transformations, lags, and parameters is a candidate hypothesis, not yet an edge. A defensible FAPP claim must report the search universe and survive selection correction, costs, a genuinely untouched holdout or independent replication, and testing outside the selected regime [31].

G. What Rice's Theorem Does and Does Not Say

Rice's theorem makes every nontrivial extensional property of arbitrary partial computable functions undecidable [32]; in particular, totality is undecidable in general [12]. It does not make a fixed total strategy's gain on a completed finite computable path undecidable, because that program can simply be simulated. Theorems 2 and 3 instead give the required promise-domain results by explicit Halting reductions: the unbounded questions of eventual profit and future-event occurrence are undecidable even though every code produced by the reductions is total.

III. THE NO-ARBITRAGE FOUNDATION

Here the market is no longer an opponent choosing a response after seeing the trader's action. Instead, financial admissibility and no-arbitrage delimit which trader-market pairs are coherent.

A. Discounted Gains and Admissibility

Let the market be defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$. Let $B_t > 0$ be a numeraire and S_t the vector of traded asset prices. Write $\bar{S}_t = S_t/B_t$ for discounted prices. A predictable, $\bar{S}$ -integrable strategy H has discounted self-financing gain

$$G_t = (H \cdot \bar{S})_t, \quad G_0 = 0. \quad (10)$$

It is *admissible* if there is a finite constant a such that

$$G_t \geq -a \quad \text{for all } t \in [0, T], \quad \mathbb{P}\text{-a.s.} \quad (11)$$

This uniform lower bound rules out doubling systems financed by unbounded credit.

Observation 7 (Universality Implies Classical Arbitrage). *If an admissible self-financing strategy H with zero initial capital satisfies*

$$G_T \geq 0 \quad \mathbb{P}\text{-a.s.}, \quad \mathbb{P}(G_T > 0) > 0,$$

then H is a classical arbitrage. In particular, a pointwise universal strategy is a classical arbitrage whenever its gain is defined on an admissible path class carrying $\mathbb{P}$.

Proof. The displayed payoff conditions, together with zero initial cost, self-financing, and admissibility, are the standard classical-arbitrage conditions. Pointwise strict positivity on every admissible path implies them, with $\mathbb{P}(G_T > 0) = 1$. $\square$

Remark 2 (Terminology and Strength). The pathwise condition is sometimes called a sure or strong arbitrage, but it is not synonymous with *arbitrage of the first kind*. The latter is the condition dual to No Unbounded Profit with Bounded Risk (NUPBR/NA1) and is a distinct, weaker viability concept [33]. Classical no-arbitrage alone already excludes the universal strategy in Observation 7; NFLVR is invoked below because it supplies the standard martingale-measure formulation.

B. NA, NUPBR, and Benchmark-Relative Gains

The relevant no-arbitrage notions do not form the simple chain sometimes suggested in informal discussions. In the standard semimartingale setting, NFLVR combines classical NA with NUPBR, whereas NA and NUPBR are in general logically distinct. NUPBR is equivalent, under standard portfolio hypotheses, to the existence of a numeraire portfolio whose positive relative-wealth processes are supermartingales [33, 34]. It may hold even when the stronger NFLVR condition and an ELMM fail.

This distinction prevents two overstatements. First, Observation 7 uses only classical NA; NUPBR alone is not silently substituted for it. If a zero-cost gain were nonnegative at all intermediate times and positive at terminal time, scaling would violate NUPBR, but terminal pathwise positivity together with only a strategy-specific lower bound does not by itself supply that scaling argument. Second, benchmark and stochastic-portfolio approaches can generate gains *relative* to a numeraire or market portfolio without producing zero-cost positive gain on every path. Relative arbitrage in diverse equity markets and real-world benchmark pricing therefore do not contradict the theorem [35, 36].

C. The Martingale Constraint

For a locally bounded semimartingale price process, the Delbaen–Schachermayer form of the First Fundamental Theorem states that No Free Lunch with Vanishing Risk (NFLVR) is equivalent to the existence of an equivalent local martingale measure (ELMM) $\mathbb{Q} \sim \mathbb{P}$ under which $\bar{S}$ is a local martingale [37, 38]. For general unbounded semimartingales the corresponding formulation uses an equivalent σ -martingale measure [39]; the corollary below retains the locally bounded setting.

Corollary 8 (No Universal Zero-Cost Strategy). *If an ELMM $\mathbb{Q}$ exists, no admissible self-financing strategy with zero initial capital can satisfy $G_T \geq 0$, $\mathbb{P}$ -almost surely, and $\mathbb{P}(G_T > 0) > 0$. Consequently, no pointwise universal strategy exists among admissible strategies in this stochastic model.*

Proof. Under $\mathbb{Q}$, the admissible stochastic integral $G = H \cdot \bar{S}$ is a local martingale bounded from below and therefore a supermartingale. Hence $\mathbb{E}_{\mathbb{Q}}[G_T] \leq G_0 = 0$. If the displayed arbitrage conditions held, equivalence would give $G_T \geq 0$, $\mathbb{Q}$ -almost surely, with $\mathbb{Q}(G_T > 0) > 0$. Consequently $\mathbb{E}_{\mathbb{Q}}[G_T] > 0$, a contradiction. $\square$

The ELMM conclusion concerns discounted gains under the pricing measure $\mathbb{Q}$. It does not say that every risky strategy has non-positive expected return under the physical measure $\mathbb{P}$; positive physical expected returns may compensate risk.

D. Win Rate Is Not Expected Value

The probability of closing a trade at a profit can greatly exceed $1/2$ even in a fair market. Optional stopping gives the precise distinction.

Proposition 9 (Asymmetric Barriers in a Fair Random Walk). *Let X_n be a simple symmetric random walk with $X_0 = 0$, and for positive integers a, b let*

$$\tau = \inf\{n \geq 0 : X_n = a \text{ or } X_n = -b\}.$$

Then

$$\mathbb{P}(X_\tau = a) = \frac{b}{a+b}, \quad \mathbb{E}[X_\tau] = 0. \quad (12)$$

Proof. For this finite interval, $\tau < \infty$ almost surely and $|X_{n \wedge \tau}| \leq \max\{a, b\}$. Optional stopping gives $\mathbb{E}[X_{n \wedge \tau}] = 0$, and bounded convergence yields $\mathbb{E}[X_\tau] = 0$ [40]. If $p = \mathbb{P}(X_\tau = a)$, then $0 = \mathbb{E}[X_\tau] = ap - b(1 - p)$, which gives $p = b/(a + b)$. $\square$

With a take-profit of $a = 1$ and a stop-loss of $b = 9$, the win probability is 0.9, but the expected payoff is zero: nine frequent unit gains are offset by one nine-unit loss in expectation. Thus prediction accuracy, trade win rate, and expected discounted profit are three different quantities.

IV. THE COMBINATORIAL PERSPECTIVE

Here the opposing side is an ensemble of market records rather than an adaptive market. The optimization NFL theorem and the elementary prediction result needed for market-direction claims should be distinguished.

Theorem 10 (Classical Optimization NFL, Wolpert—Macready). *Let X and Y be finite and give the function class Y^X its uniform law. For any two non-revisiting black-box algorithms and any fixed number $q \leq |X|$ of queries, the resulting q -term sequences of observed values have the same distribution. Consequently, every performance functional depending only on that observed value sequence has the same average for both algorithms [41].*

The theorem concerns oracle-based optimization over function spaces. It does not, by itself, imply a statement about self-financing gains on stochastic price paths. A sharpening makes its structural assumption explicit: for a uniform distribution on a finite function class, the NFL equality for all non-revisiting algorithms holds precisely when the class is closed under permutations of the search domain [42]. Financial path classes are generally not permutation-closed: temporal order, continuity, financing, and no-arbitrage restrictions distinguish one permutation from another. Thus a financial edge, when present, resides in exactly such symmetry-breaking structure and the inductive bias used to model it.

For binary directional prediction the relevant result is simpler and can be derived directly.

Proposition 11 (Uniform Binary Prediction). *Let $X_1, \dots, X_T$ be uniformly distributed on $\{0, 1\}^T$. A causal deterministic predictor chooses $\hat{X}_t = f_t(X_1, \dots, X_{t-1})$. Its mean accuracy*

$$A_T = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\hat{X}_t = X_t\}$$

satisfies $\mathbb{E}[A_T] = 1/2$. The same holds for a randomized predictor driven by a private random seed independent of $(X_1, \dots, X_T)$.

Proof. Conditional on every history $X_1, \dots, X_{t-1}$, the next bit X_t is an independent fair bit. Therefore $\mathbb{P}(\hat{X}_t = X_t \mid X_1, \dots, X_{t-1}) = 1/2$. Linearity of expectation gives the result. In the randomized case, conditioning additionally on the independent private seed leaves X_t fair given the history and proves the same equality. $\square$

Consequently, no predictor can achieve expected directional accuracy greater than $1/2$ under the uniform distribution over all binary sequences; in particular, none can exceed $1/2$ on every sequence. This is an NFL symmetry statement, not a model of actual markets. Financial sign sequences are not uniform in general, and a directional forecast does not determine trading profit because payoff sizes, financing, and stopping rules matter (Proposition 9).

The defensible practical conclusion is conditional: a claim of repeatable success must identify non-uniform structure in the physical measure $\mathbb{P}$, such as a trend, mean reversion, a risk premium, information, or an institutional constraint that breaks the NFL symmetry. The hypothesized edge concerns a restricted environment, not universal algorithmic superiority.

V. PRACTICAL ROBUSTNESS DIAGNOSTICS: TIME REVERSAL AND PERMUTATIONS

This section turns the trader–market leitmotif into a falsification tool. Hold the trader fixed and transform only the market record. Among the countable family of computably specified stress transformations, time reversal is one of the simplest, particularly for detecting dependence on directional persistence. It is not a further impossibility theorem on the level of the Halting reductions or no-arbitrage.

Let $\mathcal{M}$ be a space of strictly positive discounted price paths $P: [0, T] \rightarrow \mathbb{R}_{>0}^n$. The time reversal of P is

$$\tilde{P}(t) = P(T - t), \quad t \in [0, T]. \quad (13)$$

Literal reversal reuses the observed price levels and preserves the magnitudes of log-price moves while reversing their order and signs. It is not the same as merely negating arithmetic returns: if a forward gross return is R , the corresponding reversed gross return is $1/R$. In an empirical implementation the strategy must be rerun causally from scratch on the reversed record; reversing the original orders or realized profit would leak future information. If the initial level is prescribed, literal reversal can leave the market class. One may instead test componentwise normalized reversal,

$$P_i^{\text{rev}}(t) = P_i(0) \frac{P_i(T - t)}{P_i(T)},$$

provided it remains in the claimed class. Scaling can change the actions of a strategy that is not scale invariant; it is therefore a distinct test. For a causal strategy σ , let $\Pi_t[\sigma; P]$ denote its discounted gain from zero initial capital through time t on P . Call σ *pathwise admissible* on $\mathcal{M}$ if some finite C satisfies $\Pi_t[\sigma; P] \geq -C$ for every $P \in \mathcal{M}$ and $t \in [0, T]$.

Definition 1 (Pathwise Universal Strategy). A causal strategy σ is *pathwise universal* on $\mathcal{M}$ if it is self-financing and pathwise admissible there and satisfies $\Pi_T[\sigma; P] > 0$ for every $P \in \mathcal{M}$.

Criterion 1 (Reversal Falsification Test). Suppose $\mathcal{M}$ is closed under time reversal. If there exists $P \in \mathcal{M}$ such that

$$\min\{\Pi_T[\sigma; P], \Pi_T[\sigma; \tilde{P}]\} \leq 0, \quad (14)$$

then σ is not pathwise universal on $\mathcal{M}$.

Proof. Both P and $\tilde{P}$ belong to $\mathcal{M}$. A non-positive gain on either member of the pair contradicts Definition 1. $\square$

One non-winning path refutes a claim quantified over every path. Reversal supplies a paired candidate, not a proof that one member must fail: a causal strategy may take different positions on the reversed history. Diagonalization and no-arbitrage provide the genuine impossibility results under their respective assumptions.

a. Permuting a finite record Time reversal is a special permutation of a sampled record. For a finite record of discounted price levels $P = (P_0, \dots, P_N)$, let π be a permutation of $\{0, \dots, N\}$ and define

$$(P^\pi)_t = P_{\pi(t)}, \quad t = 0, \dots, N. \quad (15)$$

Thus $\pi(t) = N - t$ gives reversal. A fixed algorithm or a random shuffle can choose π . For vector-valued prices, each complete price vector moves together. Let $\mathcal{M}_N$ be the claimed class of such discrete records and $\Pi_N[\sigma; P]$ the discounted gain from zero initial capital when the same causal rule is rerun from its prescribed initial state on P . Use the discrete version of Definition 1 on this class.

Criterion 2 (Permutation Falsification Test). A pathwise universal strategy on $\mathcal{M}_N$ must satisfy

$$\begin{aligned} \Pi_N[\sigma; P^\pi] &> 0 \\ \text{whenever } P \in \mathcal{M}_N \text{ and } P^\pi \in \mathcal{M}_N. \end{aligned} \quad (16)$$

One permitted permutation with zero or negative gain refutes that universal claim. The criterion requires positivity, not equality of profits across permutations.

Proof. Every retained P^π is a member of the claimed market class, so strict profit on every member requires strict profit on this one. $\square$

The shuffle rearranges price levels, not returns or previously chosen orders. At each date the rerun uses only the history available in the new order. Permuting levels preserves their multiset but generally changes the returns and their magnitudes. It can also violate a fixed initial-price condition, a bound on price moves, or another restriction defining $\mathcal{M}_N$. Only transformations that remain within that class test its universal claim; closure under all permutations is unnecessary. Failure after destroying a trend outside a strategy's stated domain does not refute a claim confined to that domain. Passing finitely many shuffles of one record does not establish universal profitability on a larger path class.

Unlike the permutation closure used in optimization NFL (Section IV), this diagnostic assumes no uniform distribution or equality of average performance. Randomness in the choice of stress records alone supplies neither a market law nor a statistical significance level.

Greg Egan's *Permutation City* (1994) [43] provides a literary analogy by exploring the temporal ordering of simulated computational states. Here permutations

change the trader's observation sequence; the causal rule is rerun on each record, and Criterion 2 follows from the universal-profit requirement.

Remark 3 (Computable Stress Tests Beyond Reversal). Time reversal is one of countably many computable probes: sign reversal, regime splicing, jumps, delays, volatility rescaling, financing, and transaction costs. Since program totality is undecidable, an operational battery must select known-terminating transformations whose outputs remain in the claimed market class. Here “test” means empirical stress, not a Martin-Löf test: a uniformly computably enumerable sequence of open sets with effective measure bounds. Reversal defines no null set and certifies no randomness; passing it shows only tested robustness, not universality.

Remark 4 (Filtrations and Time Reversal). Stochastic model classes are not automatically reversal-closed: $P(T - t)$ is generally not adapted to the forward filtration, and reversing a diffusion requires additional hypotheses [44]. For example, if

$$X_t = \log(S_t/S_0) = mt + \sigma W_t,$$

then $X_{T-t} - X_T$, under its reversed filtration, has drift $-m$ and volatility σ , so a class containing both drift signs is closed in law under reversed increments. Physically, $m = \mu - \sigma^2/2$; under a risk-neutral measure the discounted stock may be a martingale while its logarithm retains Itô drift $-\sigma^2/2$. Neither an ELMM nor a driftless arithmetic-Brownian special case establishes detailed balance in general. The diagnostic remains pathwise; Section III gives the probabilistic constraints.

VI. CASE STUDY: THE WHEEL OPTIONS STRATEGY

The Wheel makes the trader–market configuration concrete. It sells a cash-secured put and, after assignment, sells a covered call against the acquired stock. For one matched pair of European options, with strike K and maturity T , the terminal-payoff identity is

$$S_T - (S_T - K)^+ = K - (K - S_T)^+ = \min\{S_T, K\}. \quad (17)$$

The cash leg pays K at maturity and must be financed at its present value; dividends and other cash flows must be matched separately. This is a local payoff comparison, not a model of the rolling Wheel. American exercise, assignment, changing strikes and maturities, roll rules, spreads, taxes, and financing make the realized strategy path dependent. Each short-option leg nevertheless carries the associated downside exposure.

Option selling can nevertheless have positive mean return under a physical law. The difference between risk-neutral and physical expectations of future variance is commonly called the variance risk premium, with sign conventions varying across the literature, and option-market evidence documents compensation for bearing

volatility and jump risk [45, 46]. Such compensation is empirical and state-dependent: its sign and magnitude depend on option prices, strikes, transaction costs, and the physical distribution.

The local payoff comparison suggests four stress scenarios for the rolling strategy. A crash after assignment produces essentially one-for-one downside below K . A persistent decline can make repeated premiums too small to offset the renewed short-put exposure. In a large rally, assignment and call exercise can still leave a positive absolute profit, but the cap at K can underperform a buy-and-hold benchmark. Finally, capital exhaustion is absorbing for a self-financing trader: if $W_{t+1} = W_t(1 + fR_{t+1})$ and $1 + fR_{t+1} = 0$, then $W_{t+1} = 0$ unless external capital is injected. A fully cash-secured, unlevered implementation has bounded loss, but an underlying that falls to zero can consume almost all committed capital net of premium.

A rally/crash reversal is therefore a useful paired stress test, although the payoff identity, rather than reversal alone, supplies the economic mechanism. Repeated small premiums also need not imply satisfactory time-average growth:

$$\frac{1}{T} \log \frac{W_T}{W_0} = \frac{1}{T} \sum_{t=1}^T \log(1 + fR_t),$$

so frequent gains can coexist with poor long-run growth when rare losses are large [47]. The Wheel may be a defensible FAPP strategy under a specified risk premium, capital policy, cost model, and benchmark; it is not a pathwise profit guarantee.

VII. UNIVERSAL PORTFOLIOS AND DISTANCE FROM UNIVERSALITY

Here the trader changes the claim made against the market side: it seeks small regret to a benchmark portfolio class, not positive profit on every path. The word “universal” also appears in a celebrated positive theorem. For price-relative vectors $x_t \in \mathbb{R}_{>0}^m$, a constant-rebalanced portfolio b has wealth

$$S_T(b) = \prod_{t=1}^T b^\top x_t.$$

Cover's universal portfolio is nonanticipating and, under the conditions of his theorem, has the same asymptotic exponential growth rate as the best constant-rebalanced portfolio chosen in hindsight on every individual bounded sequence [48]. In regret notation,

$$\frac{1}{T} \left[\log \max_b S_T(b) - \log S_T^{\text{UP}} \right] \longrightarrow 0. \quad (18)$$

Equation (18) is a relative guarantee, not an absolute-profit guarantee. If every constant-rebalanced portfolio loses money on a path, the universal portfolio may lose as

well while retaining vanishing per-period regret. Cover’s result therefore does not contradict Corollary 8; it illustrates a useful benchmark-relative meaning of “universal” once the comparison class is specified.

Online regret methods also yield financial guarantees of another kind. DeMarzo, Kremer, and Mansour show that gradient strategies designed to minimize asymptotic regret imply trading strategies that provide arbitrage-based option price bounds without assuming continuous price paths, complete markets, or a pricing kernel; the bounds depend on realized quadratic variation [49]. This is robust pricing and hedging, not guaranteed positive profit or next-move prediction.

a. A Hierarchy of Attainable Claims The useful alternatives to pathwise universality are not weaker points on one single order; they change the benchmark, horizon, or probability quantifier. Table II records the distinctions.

Superhedging duality gives another quantitative notion: once a contingent claim and an admissible model class are fixed, its superhedging price is the least initial capital needed for a pathwise guarantee. Requiring zero initial capital while demanding a strictly positive payoff is precisely the degenerate, arbitrage-producing endpoint ruled out above.

VIII. FAPP STRATEGIES AND THEIR HONEST SCOPE

Although pathwise universal profit is excluded, a strategy may be profitable *for all practical purposes* (FAPP, following Bell [50]) relative to a physical law $\mathbb{P}$, a benchmark, and a finite horizon. Market making may earn spreads when adverse selection and inventory risk are controlled, and Kelly investing maximizes expected logarithmic growth when the relevant distribution is sufficiently well specified [51]. Neither is an unconditional guarantee.

Here “insight” means a falsifiable restriction on the market side. In actual trading it usually appears as a heuristic: persistence, mean reversion, valuation correction, or compensation for an identified risk. The fact that a heuristic is algorithmic does not disqualify it; its edge is a regime-dependent empirical hypothesis. A repeatable ex ante claim must therefore specify its market class, horizon, costs, benchmark, and validation protocol.

Only lawfully acquired and used information is included in this practical strategy class; an informational advantage remains short of a pathwise guarantee.

The unit of performance matters as well. For a funded portfolio, let R^{nom} be nominal total return and π inflation over the same interval under a stated price index. Then

$$1 + R^{\text{real}} = \frac{1 + R^{\text{nom}}}{1 + \pi}. \quad (19)$$

A positive nominal return can therefore be a purchasing-power loss. The price index is an economic deflator, not

necessarily a tradable numeraire for the FTAP; the consumption benchmark and financing account serve different purposes.

Productivity growth may support positive real returns when its benefits reach investors as cash flows [52]. Prices, discount rates, and the displacement of incumbents intervene [53]; aggregate growth therefore establishes neither next-move prediction nor active alpha.

Equilibrium theory likewise permits only conditional edges. Grossman and Stiglitz show that perfectly informationally efficient markets are inconsistent with costly information acquisition: if prices revealed all information without reward, agents would not pay to produce it [54]. Conversely, under its common-prior, common-knowledge, risk-aversion, and Pareto-optimality assumptions, the Milgrom–Stokey no-trade theorem limits trade based on private information alone [55]. A complementary market-selection result is also conditional: among agents in Sandroni’s complete-market model who share an intertemporal discount factor and choose saving endogenously, accurate forecasters prosper and inaccurate ones are driven out [56]. These equilibrium results depend on their model assumptions; none is a pathwise trading guarantee or a substitute for the FTAP.

Deployment can also change the market side. A large, crowded, or public strategy consumes liquidity, moves quotes, attracts imitation or opposition, and changes the distribution on which it was estimated. Lo’s Adaptive Markets Hypothesis provides this evolutionary interpretation [57]: competition and impact may erode an edge without producing the tailored countermarket of Theorem 1. Feedback invalidates scale-free extrapolation; it does not imply that every large strategy loses.

Finally, the finite-pattern trap of Section IIF becomes operational in strategy discovery. Trying many specifications and retaining the best backtest turns its Sharpe ratio into an order statistic. The Deflated Sharpe Ratio corrects for selection, non-normality, and the number of trials [31]; untouched validation, costs, and deployment analysis remain necessary. These controls do not prove an edge, but make a finite-sample FAPP claim more honestly falsifiable.

IX. CONCLUSION

Everything returns to the same pair: a trader and a market. For every fixed total deterministic computable trader, Theorem 1 constructs a fixed positive computable price path whose next bounded proportional move opposes every nonzero position and remains nonzero during inactivity. The quantifiers are $\forall \sigma \exists \mathcal{E}_\sigma$: no physical market must observe the deployed orders, the always-flat trader earns zero, and every nonzero position loses on its trader-relative countermarket. This direct diagonal result uses no Halting-Problem reduction and identifies no single path that defeats every trader.

The remaining limits concern different market sides.

TABLE II. Different meanings of a successful or universal strategy.

Notion	Guarantee	Status
Pathwise universal profit	Positive discounted zero-cost gain on every admissible path	Excluded among admissible strategies in the specified stochastic model under classical NA
Relative arbitrage	Outperformance of a specified market or numeraire portfolio	Possible under structural market assumptions
Cover universality	Vanishing per-period log regret to the best constant-rebalanced portfolio	Pathwise but benchmark-relative
FAPP or statistical edge	Positive performance under a specified physical law, data protocol, and horizon	Conditional and empirically falsifiable

Gold precludes identifying every computable binary generator in the limit, while next-bit diagonalization defeats each fixed computable predictor on its predictor-relative record. Halting reductions preclude uniform certification of eventual profit or encoded future events, and Busy Beaver excludes a computable description-size waiting horizon despite finitely solvable low-complexity cases. Under a stated computable reference measure, the generated record is Martin-Löf random almost surely; neither this property nor Borel normality can be certified from a finite sample. Ramsey regularity explains why finite structured subsequences nevertheless remain abundant, including in random records. No-arbitrage and uniform binary averaging supply still different boundaries. Their conclusions are complementary because their inputs, quantifiers, and success criteria differ.

Positive results arise only after restricting the claim. The Wheel may earn a risk premium while retaining put-like downside; Cover guarantees asymptotic performance relative to a benchmark class; and FAPP strategies may exploit a specified regime, lawful information, or compensated risk. Such performance must be evaluated after costs and financing, against an explicit benchmark and, where relevant, in purchasing-power terms. Productivity-supported real growth is one possible regime hypothesis, not a next-move predictor or a guarantee of active alpha.

Any repeatable-success claim must therefore state its market restriction, probability law, benchmark, admissibility and financing conditions, horizon, data protocol, and deployment effects. Time reversal and other within-class permutations are robustness probes: one nonpositive gain refutes an all-path claim, while passing a finite set of probes does not certify universality. Useful strategies may exist, but no total deterministic computable self-financing rule guarantees strict gain on every positive computable price path; under an arbitrage-free stochastic model, no admissible zero-cost rule delivers nonnegative discounted gain almost surely and positive gain with positive probability.

ACKNOWLEDGMENTS

This research was funded in whole or in part by the Austrian Science Fund (FWF) [Grant DOI: 10.55776/PIN5424624]. For open access purposes, the author has applied a CC BY public copyright license to any author accepted manuscript version arising from this submission. The author acknowledges TU Wien Bibliothek for financial support through its Open Access Funding Programme.

OpenAI Codex was used to assist with literature organization, LaTeX editing, and checks of derivations. The author specified the scientific direction and prompts; model output retained in the manuscript was checked against the cited sources and explicit calculations. The author accepts full responsibility for the final text.

DATA AVAILABILITY

This is a purely mathematical work, and no data or software were created or analyzed in this study.

-
- [1] E. M. Gold, Limiting recursion, *The Journal of Symbolic Logic* **30**, 28 (1965).
 - [2] E. M. Gold, Language identification in the limit, *Information and Control* **10**, 447 (1967).
 - [3] L. Bachelier, Théorie de la spéculation, *Annales Scientifiques de l'École Normale Supérieure* **17**, 21 (1900).
 - [4] P. A. Samuelson, Proof that properly anticipated prices fluctuate randomly, *Industrial Management Review* **6**, 41 (1965).
 - [5] E. F. Fama, Efficient capital markets: A review of theory and empirical work, *The Journal of Finance* **25**, 383 (1970).
 - [6] S. Legg, *Is there an Elegant Universal Theory of Prediction?*, Tech. Rep. IDSIA-12-06 (Istituto Dalle Molle di Studi sull'Intelligenza Artificiale, 2006) [arXiv:cs/0606070 \[cs.AI\]](https://arxiv.org/abs/cs/0606070).
 - [7] R. J. Solomonoff, A formal theory of inductive inference. part I, *Information and Control* **7**, 1 (1964).
 - [8] R. J. Solomonoff, A formal theory of inductive inference. part II, *Information and Control* **7**, 224 (1964).
 - [9] R. J. Solomonoff, Complexity-based induction systems: Comparisons and convergence theorems, *IEEE Transactions on Information Theory* **24**, 422 (1978).
 - [10] M. Hutter, Convergence and error bounds for universal prediction of nonbinary sequences, in *Machine Learning: ECML 2001*, Lecture Notes in Computer Science, Vol. 2167, edited by L. De Raedt and P. Flach (Springer, 2001) pp. 239–250.
 - [11] V. V. V'yugin, On impossibility of sequential algorithmic forecasting, in *Kolmogorov Complexity and Applications*, Dagstuhl Seminar Proceedings, Vol. 6051, edited by M. Hutter, W. Merkle, and P. M. B. Vitányi (Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2006) pp. 1–7.
 - [12] A. M. Turing, On computable numbers, with an application to the Entscheidungsproblem, *Proceedings of the London Mathematical Society* **2**, 42, 230 (1937).
 - [13] L. Blum and M. Blum, Toward a mathematical theory of inductive inference, *Information and Control* **28**, 125 (1975).
 - [14] D. Angluin and C. H. Smith, Inductive inference: Theory and methods, *ACM Computing Surveys* **15**, 237 (1983).
 - [15] L. M. Adleman and M. Blum, Inductive inference and unsolvability, *The Journal of Symbolic Logic* **56**, 891 (1991).
 - [16] A. Svozil and K. Svozil, Induction and physical theory formation as well as universal computation by machine learning, in *Swarm Intelligence: From Social Bacteria to Humans*, edited by A. Schumann (CRC Press, Taylor & Francis Group, LLC, Boca Raton, FL, 2021) pp. 170–180, [arXiv:1609.03862](https://arxiv.org/abs/1609.03862).
 - [17] C. D. Moore, Unpredictability and undecidability in dynamical systems, *Physical Review Letters* **64**, 2354 (1990), cf. Ch. Bennett, *Nature*, **346**, 606 (1990).
 - [18] T. Rado, On non-computable functions, *The Bell System Technical Journal* **XLI(41)**, 877 (1962).
 - [19] G. J. Chaitin, Computing the busy beaver function, in *Open Problems in Communication and Computation*, edited by T. M. Cover and B. Gopinath (Springer, New York, 1987) p. 108.
 - [20] C. S. Calude, M. J. Dinneen, and C.-K. Shu, Computing a glimpse of randomness, *Experimental Mathematics* **11**, 361 (2002), [arXiv:nlin/0112022](https://arxiv.org/abs/nlin/0112022).
 - [21] M. Barnsley, *Fractals Everywhere* (Academic Press, San Diego, 1988).
 - [22] C. S. Calude, E. Calude, and K. Svozil, The complexity of proving chaoticity and the Church–Turing thesis, *Chaos: An Interdisciplinary Journal of Nonlinear Science* **20**, 037103 (2010).
 - [23] D. A. Martin, Borel determinacy, *Annals of Mathematics* **102**, 363 (1975).
 - [24] P. Martin-Löf, The definition of random sequences, *Information and Control* **9**, 602 (1966).
 - [25] D. G. Champernowne, The construction of decimals normal in the scale of ten, *Journal of the London Mathematical Society* **s1-8**, 254 (1933).
 - [26] R. L. Graham, B. L. Rothschild, and J. H. Spencer, *Ramsey Theory*, 2nd ed. (John Wiley & Sons, New York, 1990).
 - [27] R. Graham and J. H. Spencer, Ramsey theory, *Scientific American* **263**, 112 (1990).
 - [28] C. S. Calude, *Information and Randomness—An Algorithmic Perspective*, 2nd ed. (Springer, Berlin, 2002).
 - [29] A. Nies, *Computability and Randomness*, Oxford Logic Guides, Vol. 51 (Oxford University Press, Oxford, 2009).
 - [30] C. S. Calude and G. Longo, The deluge of spurious correlations in big data, *Foundations of Science* **22**, 595 (2017), first published online 7 March 2016, [CDMTCS-488](https://doi.org/10.1007/s11464-016-0488-8).
 - [31] D. H. Bailey and M. López de Prado, The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality, *The Journal of Portfolio Management* **40**, 94 (2014).
 - [32] H. G. Rice, Classes of recursively enumerable sets and their decision problems, *Transactions of the American Mathematical Society* **74**, 358 (1953).
 - [33] C. Kardaras, Market viability via absence of arbitrage of the first kind, *Finance and Stochastics* **16**, 651 (2012).
 - [34] I. Karatzas and C. Kardaras, The numéraire portfolio in semimartingale financial models, *Finance and Stochastics* **11**, 447 (2007).
 - [35] R. Fernholz, I. Karatzas, and C. Kardaras, Diversity and relative arbitrage in equity markets, *Finance and Stochastics* **9**, 1 (2005).
 - [36] E. Platen and D. Heath, *A Benchmark Approach to Quantitative Finance*, Springer Finance (Springer, Berlin, 2010).
 - [37] J. M. Harrison and S. R. Pliska, Martingales and stochastic integrals in the theory of continuous trading, *Stochastic Processes and their Applications* **11**, 215 (1981).
 - [38] F. Delbaen and W. Schachermayer, A general version of the fundamental theorem of asset pricing, *Mathematische Annalen* **300**, 463 (1994).
 - [39] F. Delbaen and W. Schachermayer, The fundamental theorem of asset pricing for unbounded stochastic processes, *Mathematische Annalen* **312**, 215 (1998).
 - [40] J. L. Doob, *Stochastic Processes* (John Wiley & Sons, New York, 1953).
 - [41] D. H. Wolpert and W. G. Macready, No free lunch theorems for optimization, *IEEE Transactions on Evolutionary Computation* **1**, 67 (1997).
 - [42] C. Schumacher, M. D. Vose, and L. D. Whitley, The no free lunch and problem description length, in *Proceedings of the Genetic and Evolutionary Computation Conference*

- (Morgan Kaufmann, San Francisco, 2001) pp. 565–570.
- [43] G. Egan, *Permutation City* (Orion/Millennium, London, 1994).
 - [44] U. G. Haussmann and É. Pardoux, Time reversal of diffusions, *The Annals of Probability* **14**, 1188 (1986).
 - [45] G. Bakshi and N. Kapadia, Delta-hedged gains and the negative market volatility risk premium, *The Review of Financial Studies* **16**, 527 (2003).
 - [46] P. Carr and L. Wu, Variance risk premiums, *The Review of Financial Studies* **22**, 1311 (2009).
 - [47] O. Peters, The ergodicity problem in economics, *Nature Physics* **15**, 1216 (2019).
 - [48] T. M. Cover, Universal portfolios, *Mathematical Finance* **1**, 1 (1991).
 - [49] P. M. DeMarzo, I. Kremer, and Y. Mansour, Robust option pricing: Hannan and Blackwell meet Black and Scholes, *Journal of Economic Theory* **163**, 410 (2016).
 - [50] J. S. Bell, Against ‘measurement’, *Physics World* **3**, 33 (1990).
 - [51] J. L. Kelly, A new interpretation of information rate, *The Bell System Technical Journal* **35**, 917 (1956).
 - [52] R. M. Solow, Technical change and the aggregate production function, *The Review of Economics and Statistics* **39**, 312 (1957).
 - [53] J. A. Schumpeter, *Capitalism, Socialism and Democracy* (Harper and Brothers, New York, NY, 1942).
 - [54] S. J. Grossman and J. E. Stiglitz, On the impossibility of informationally efficient markets, *The American Economic Review* **70**, 393 (1980).
 - [55] P. Milgrom and N. Stokey, Information, trade and common knowledge, *Journal of Economic Theory* **26**, 17 (1982).
 - [56] A. Sandroni, Do markets favor agents able to make accurate predictions?, *Econometrica* **68**, 1303 (2000).
 - [57] A. W. Lo, The adaptive markets hypothesis: Market efficiency from an evolutionary perspective, *The Journal of Portfolio Management* **30**, 15 (2004).